SleepTwin

A sleep-driven digital twin that forecasts tomorrow's blood-sugar spike and morning BP surge for people with Type 2 Diabetes and Hypertension

Team WeBareBears · PCCOE · Happiest Health Digital Twin Challenge 2026 - Phase 1

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Model-based projection for clinical decision support. Research prototype.

The problem: A Blind Spot between visits

~101 M

people with diabetes in India

~315 M

people with hypertension

~15 min

with a doctor every few months

~3 months of nights the doctor never sees



SleepTwin reads every night and forecasts tomorrow - every morning

Source: ICMR-INDIAB, Lancet Diabetes & Endocrinology 2023

Why sleep? Two paths from a bad night to tomorrow's risk



4 nights of 5 h → insulin sensitivity −25% (Spiegel 1999, Buxton 2010, Donga 2010)



Most sleep apnea is never diagnosed - a normal sleep app calls the night 'fine'

Sleep is measured every night by consumer wearables - and it can be changed tonight.

Our solution: A Digital Twin of each Patient



Two data streams in - the health record + daily wearable data - a personal forecast, an early warning, a reason and a tested fix out.

How SleepTwin works

From open data to a doctor's decision - left to right in four stages

1 OPEN DATA

real & anonymized - for calibration

CDC NHANES 2017-2020

Real anonymized US adults → realistic, correlated health records

LifeSnaps (Fitbit)

Real volunteers' sleep, heart-rate & HRV patterns

Published science

Bergman glucose model, sleep → insulin effect, gene frequencies

calibrates →

2 VIRTUAL PATIENTS

the simulator - 300 patients × 21 days

Health record

Age, diagnoses + years, labs, medicines, genetic markers

Wearable stream

Sleep stages, HRV, heart rate, steps, oxygen dips

Body model

Sleep debt → insulin sensitivity → glucose every 5 min; morning BP

daily data →

3 TWIN BRAIN

AI / machine learning

Learn the person's normal

Days 1-14 set a personal baseline

Predict (LightGBM)

Tomorrow's glucose spike & BP surge, plus a glucose alert 2 hours ahead

Explain + What-If

SHAP → reasons in plain English; tests a fix in ~15 ms

risk + why →

4 DOCTOR DASHBOARD

website - laptop, tablet or phone

FastAPI service

Secure API that serves risks, reasons and trends

5 pages

Patients • Overview • Trends • What-If • How It Works

Doctor decides

Decision support only - never a diagnosis

✓ Checked on real data

In 38 real Fitbit users, short nights pushed resting heart rate up and HRV down versus each person's own normal (the idea the twin is built on).

What-If loop: the doctor tests a fix (sleep longer, earlier bedtime, CPAP) → the twin re-runs the body model + AI and shows the new risk

All patients are SIMULATED • Model-based projection for clinical decision support. Research prototype. • Team WeBareBears · PCCOE

Two Data Streams + Sandbox compliance

Static: health record

- ✓ Demographics (age, sex, BMI)
- ✓ Diagnoses with years
- ✓ Lab results - latest + prior
- ✓ Medicines
- ✓ Genetic markers (simulated)

Dynamic: wearables

- ✓ Sleep stages every 5 min
- ✓ HRV + resting heart rate
- ✓ Steps
- ✓ Night-time oxygen dips
- ✓ CGM glucose every 5 min

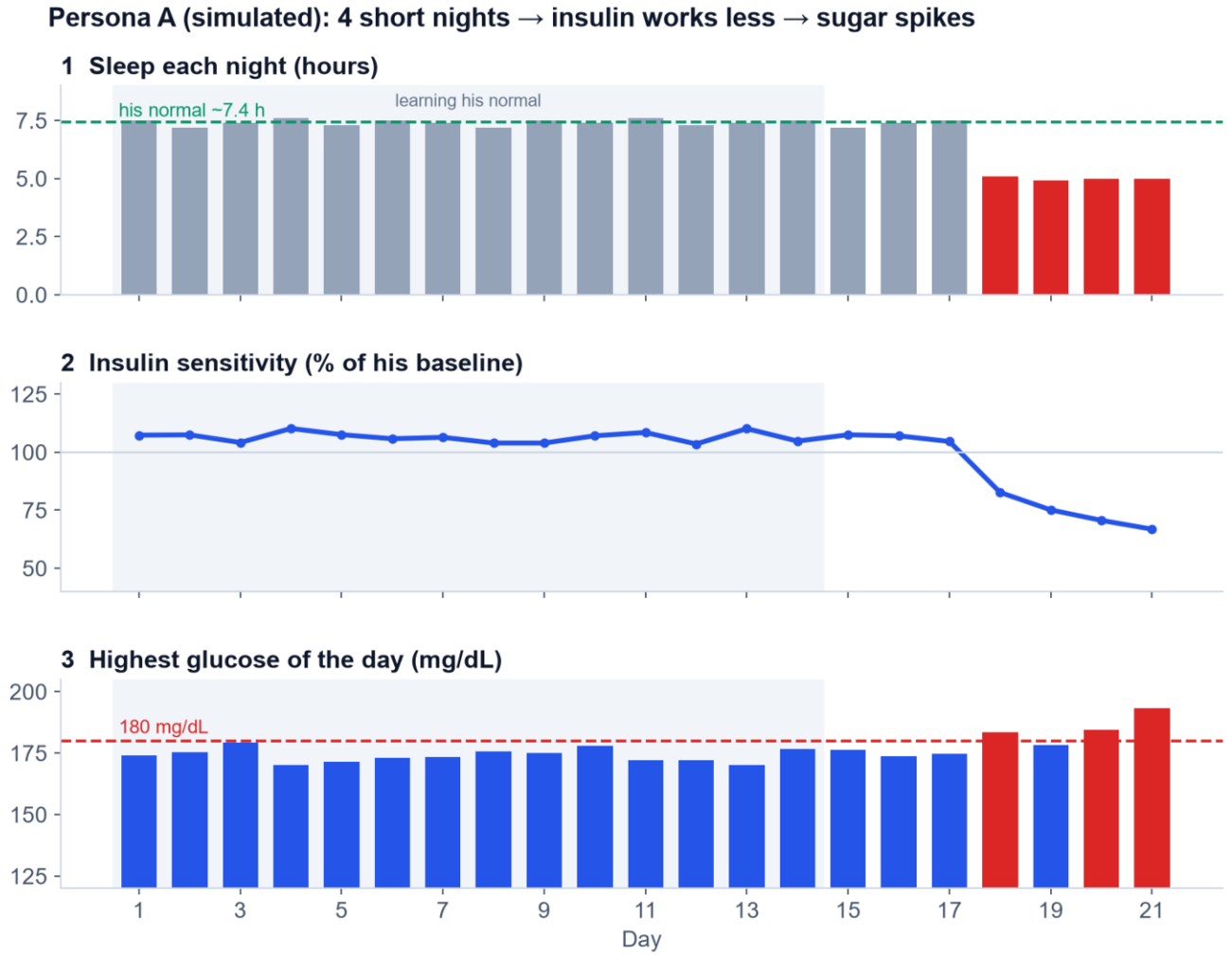
Safe data only

- ✓ 300 virtual patients × 21 days
- ✓ Calibrated on NHANES (public, anonymized)
- ✓ Patterns from LifeSnaps (open, CC-BY-4.0)
- ✓ No real patients, no personal data
- ✓ Labelled 'Simulated' everywhere

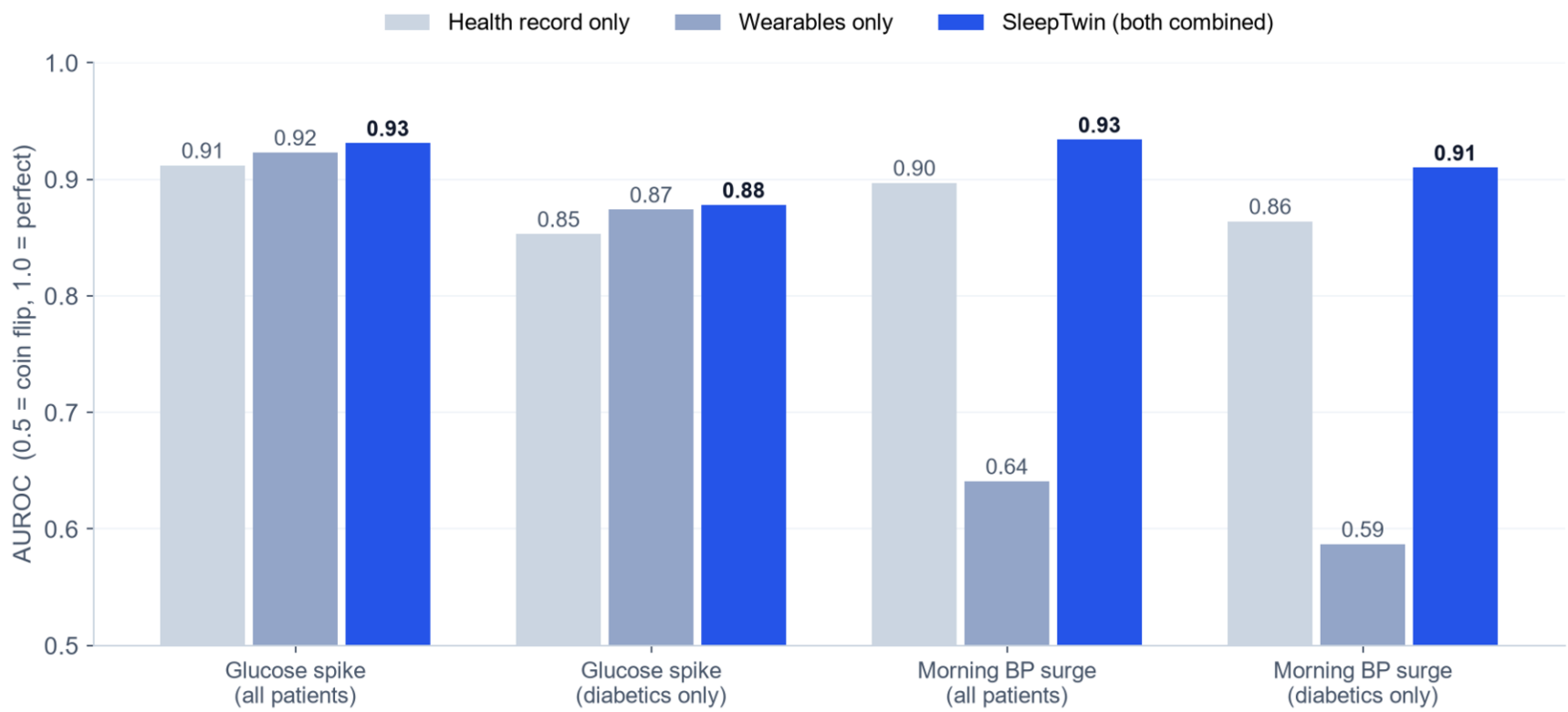
Cohort: T2D 37% · prediabetes 29% · hypertension 52% · sleep apnea 26% (only 5% diagnosed)

The simulator: How sleep reaches Blood Sugar

- Sleep debt = last 7 nights' shortfall, recent nights count more
- 4 nights of 5 h → insulin sensitivity −25% (matches sleep studies)
- Bergman glucose model turns meals + insulin sensitivity into CGM every 5 min
- Beta-blockers hide the heart-rate signal - HRV still shows the strain
- Honest limit: labels come from the same simulator



Does combining both data streams help? Yes.



2,100 patient-days, tested only on patients the model never saw (GroupKFold by patient). The gain is largest for diabetics - the same night means more for a high-risk patient.

2-hour early warning for Glucose spikes

70 min

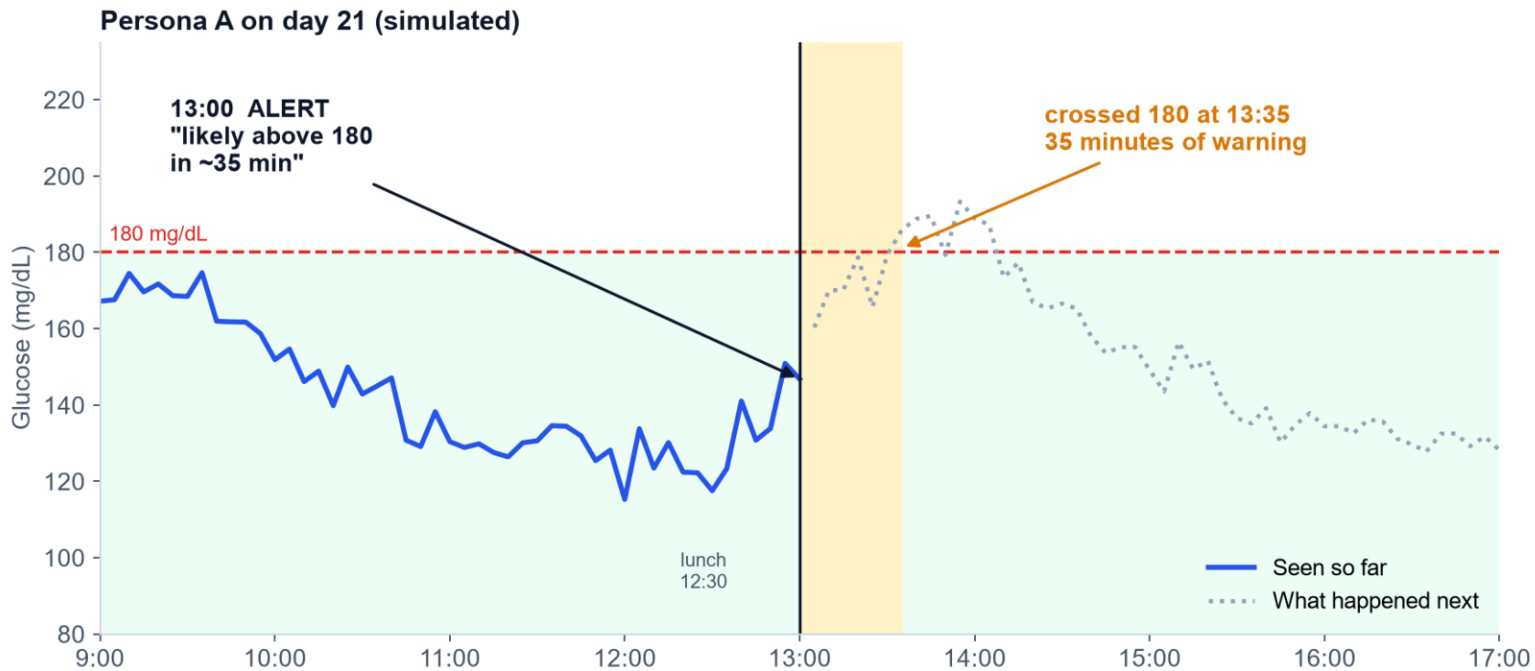
median warning before a spike

0.95

accuracy (AUROC) vs 0.79 for a plain trend arrow

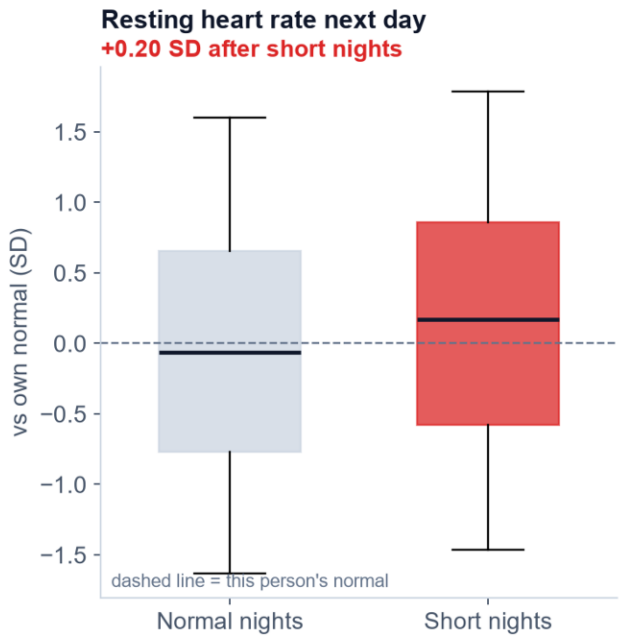
60%

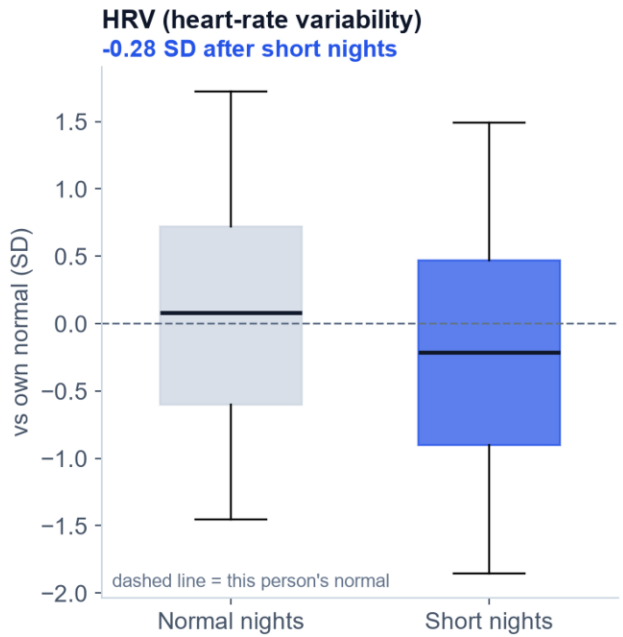
of spikes warned in advance



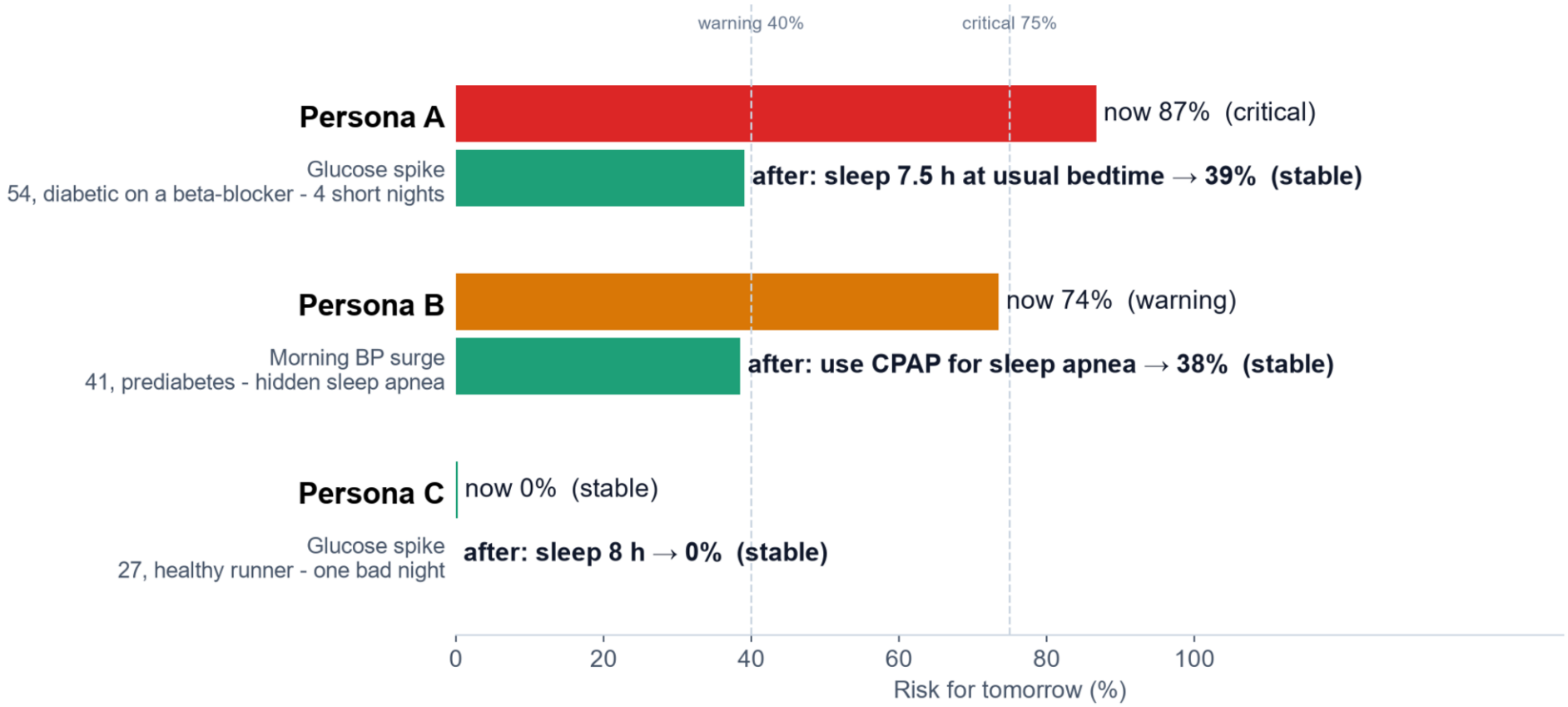
Checked on real people (LifeSnaps Fitbit data)

- 38 real users, 2,381 nights
- Short night = below that person's own usual
- Next-day resting HR: +0.20 SD (p = 0.0028)
- HRV: -0.28 SD (p = 2.4e-05)
- The twin's core idea - drift from your own normal - holds in real data



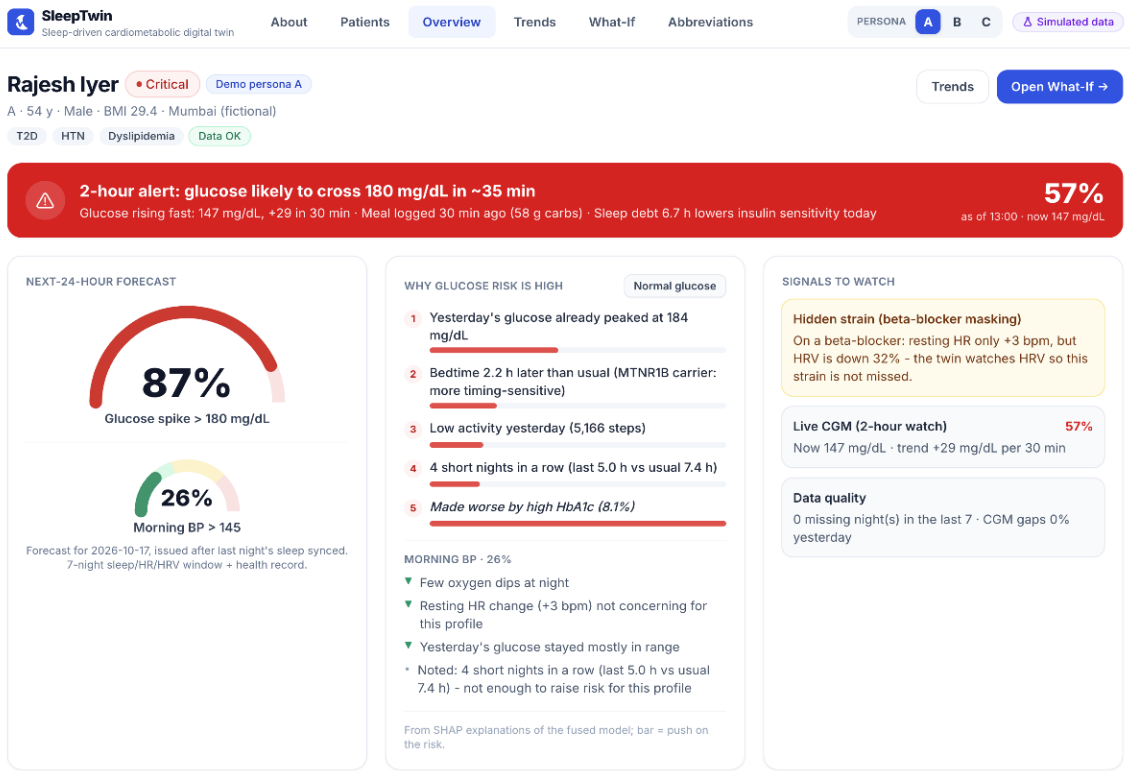


Outcome: Three Patients, Three Different Answers

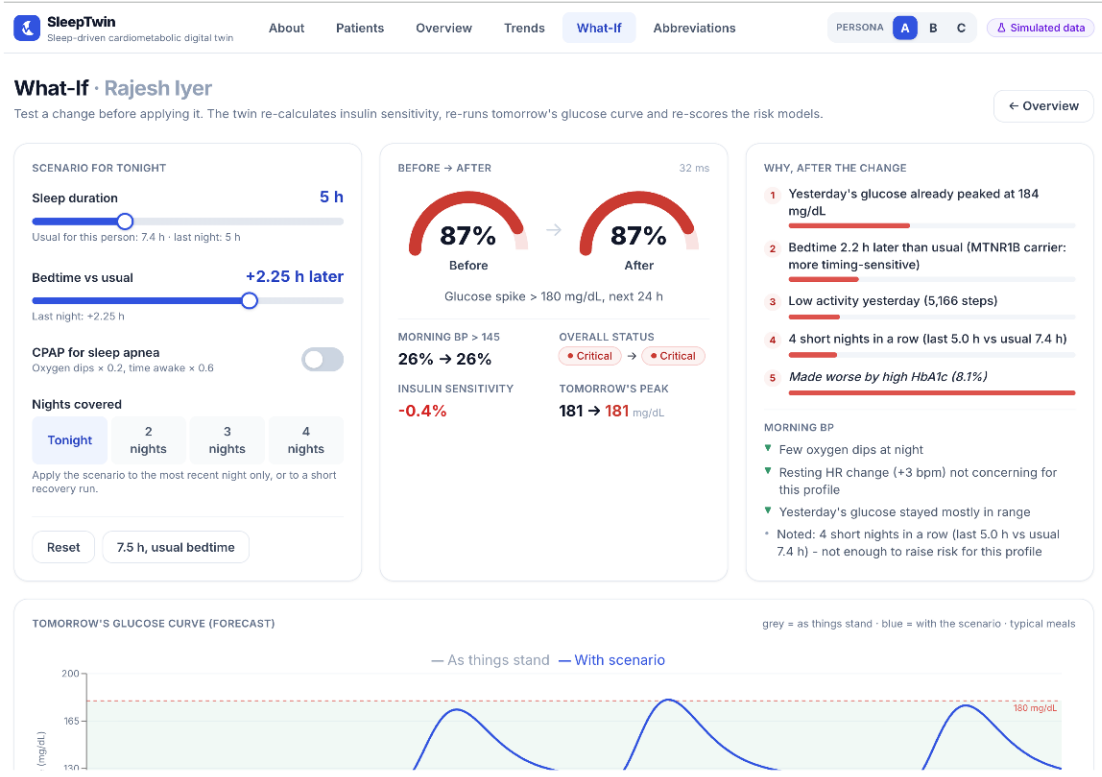


The twin escalates the high-risk patient, finds a problem a sleep app would miss, and stays calm for a healthy person.

The Clinician Dashboard



Overview: risk, 2-hour alert, reasons, hidden-signal flags



What-If: move the sleep slider - risk and tomorrow's curve update

Limitations & Safety

Decision support only

- ✓ Not a diagnosis
- ✓ The doctor stays in charge
- ✓ Disclaimer on every screen

Not clinically proven

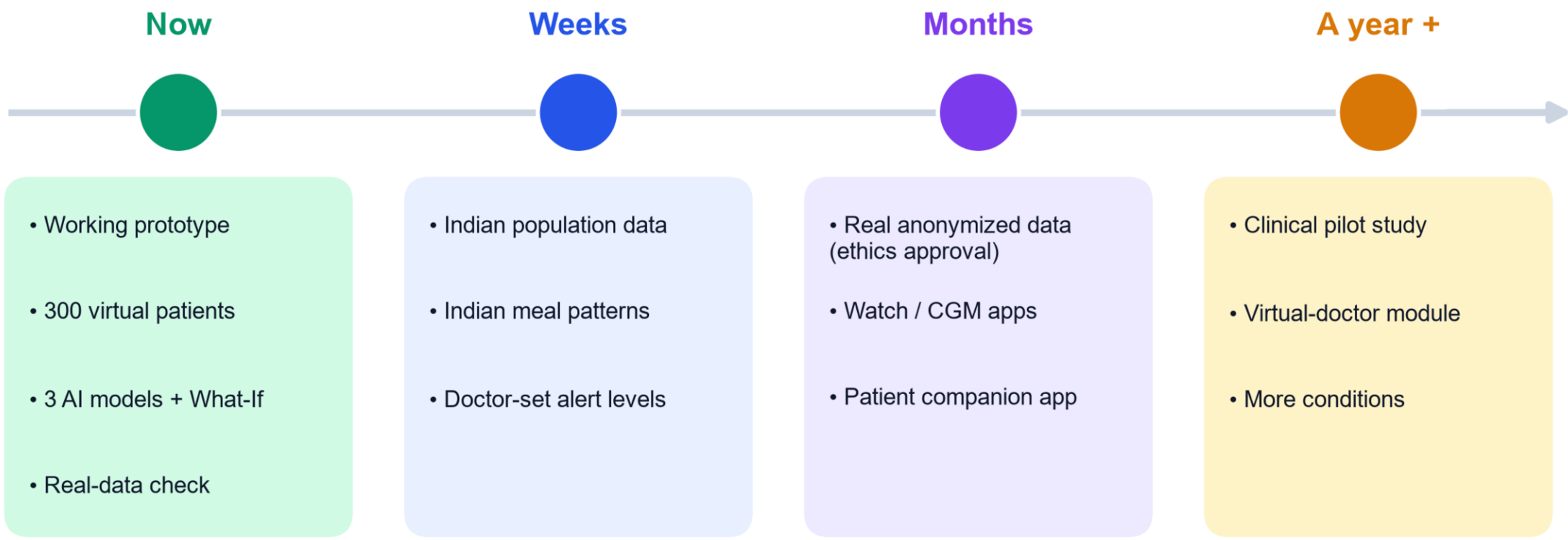
- ✓ Trained on virtual patients
- ✓ Labels share the simulator's assumptions
- ✓ Needs real-data validation

Built safely

- ✓ No real patient data
- ✓ Security-hardened app + API
- ✓ 0 known dependency vulnerabilities

Also: NHANES is a US population and genetic markers are simulated - Indian calibration is future work.

Roadmap & Fit with a Virtual Doctor Programme



SleepTwin can be the between-visit 'sleep + metabolism' module of a virtual doctor: forecast tomorrow, suggest tonight's fix.